



Databases and Applications

Unit III

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Computer Applications

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Document Models: JSON, BSON, Document DB Basics

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► What is a Document Model?

- Stores data as documents instead of tables
- Each document groups related fields together
- Represents real-world entities in a natural way

► Why is it used?

- Extremely flexible and schema-less structure
- Ideal for nested, hierarchical data
- Widely used in modern web and mobile applications



► What is JSON?

- JavaScript Object Notation (lightweight format)
- Stores data as key-value pairs
- Human-readable and easy to exchange between systems

► JSON Structure

- Objects enclosed in { }
- Arrays enclosed in []
- Supports strings, numbers, booleans, arrays, and nested objects



Databases and Applications

JSON – Example

► Simple JSON Document

```
{  
  "name":  "Aarathi",  
  "age":   23,  
  "skills": ["Python", "SQL"]  
}
```

► Why JSON is useful

- Clean way to represent a single entity
- Handles lists and nested fields easily
- Used in REST APIs, NoSQL databases, and configuration files



► What is BSON?

- Binary JSON format used internally by MongoDB
- Designed for fast reading and writing
- Supports additional types like Date, ObjectId, Binary

► Why not JSON alone?

- JSON lacks complex data types
- JSON is text-based → slower to parse
- BSON provides type metadata and indexing advantages



JSON vs BSON & Document Databases

► JSON vs BSON

- JSON → human-readable, lightweight, text format
- BSON → machine-readable binary, larger but faster
- BSON enables efficient database queries and indexes

► Document Databases

- Store information as JSON/BSON documents
- Schema-less: structure may vary between documents
- Examples: MongoDB, CouchDB, Firebase Firestore



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